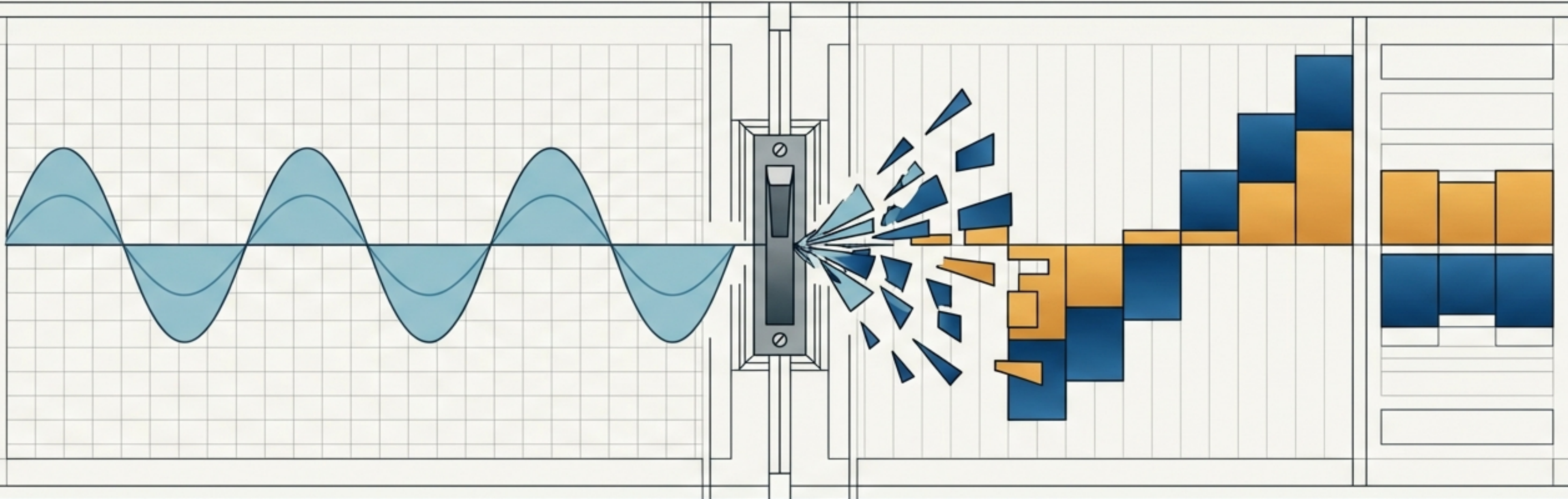


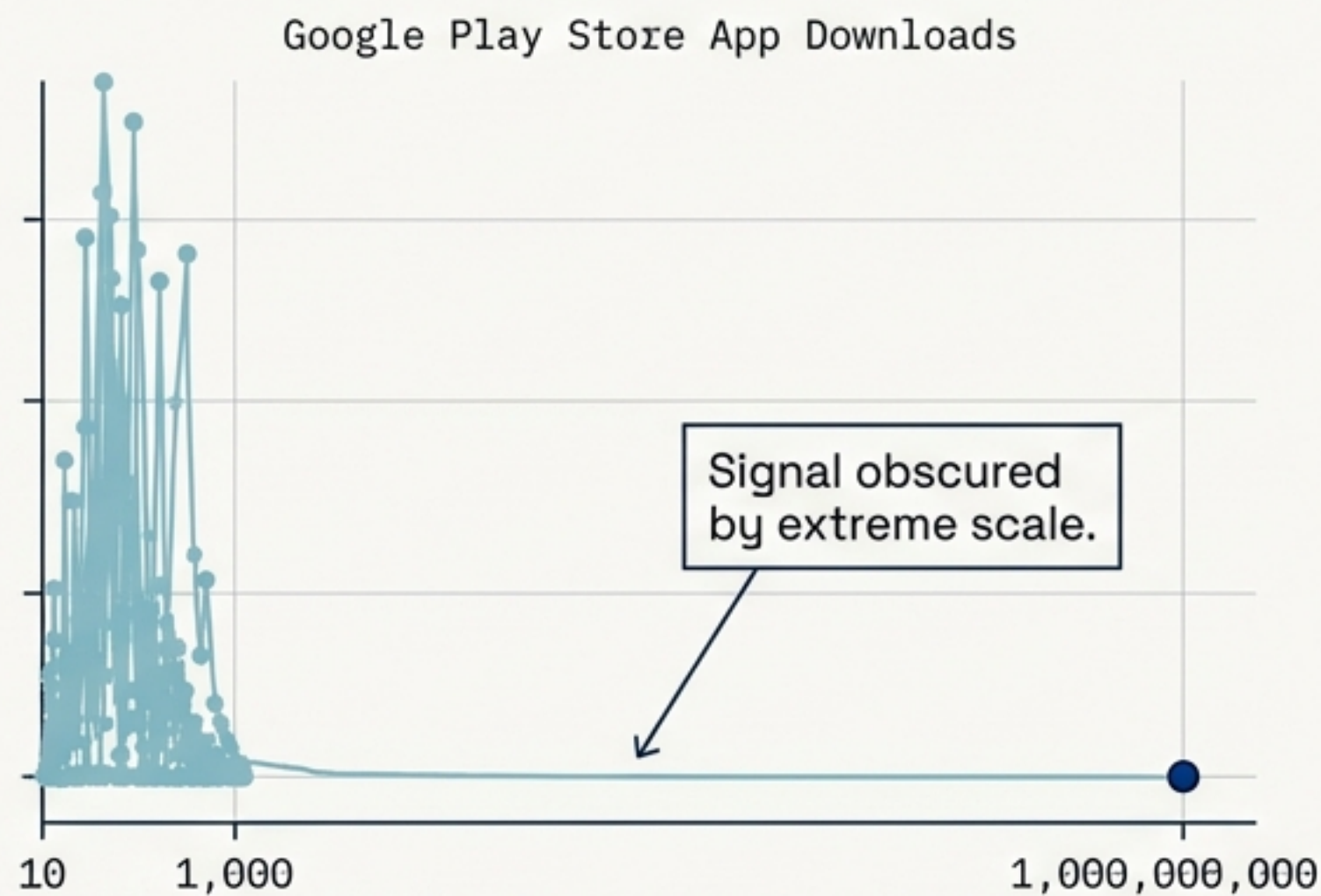
Encoding Numerical Data: A Masterclass in Feature Engineering



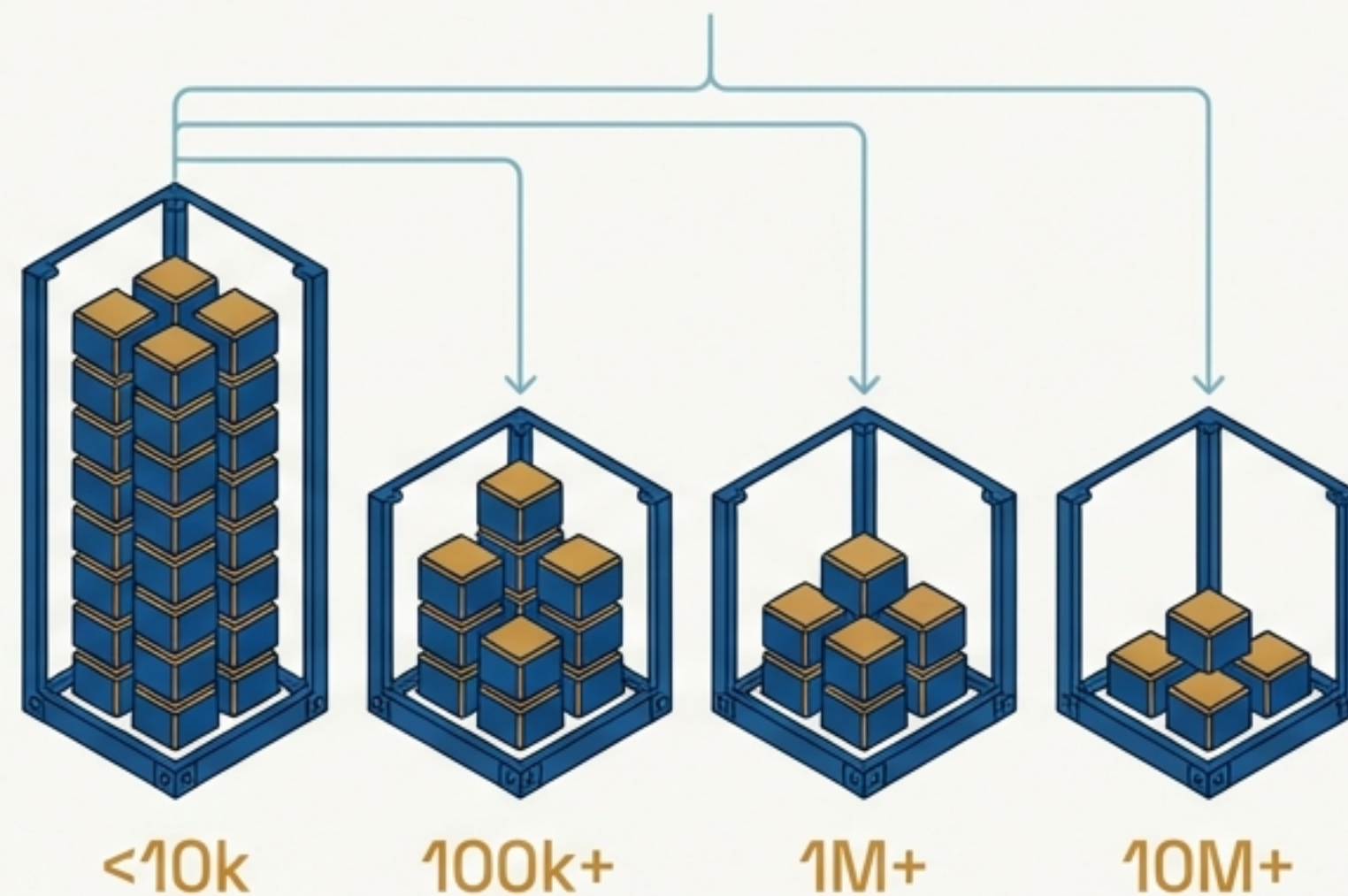
Discretisation, Binning, and Binarisation Techniques

The Core Problem: Why Touch Numerical Data?

The Mess: Continuous Raw Data

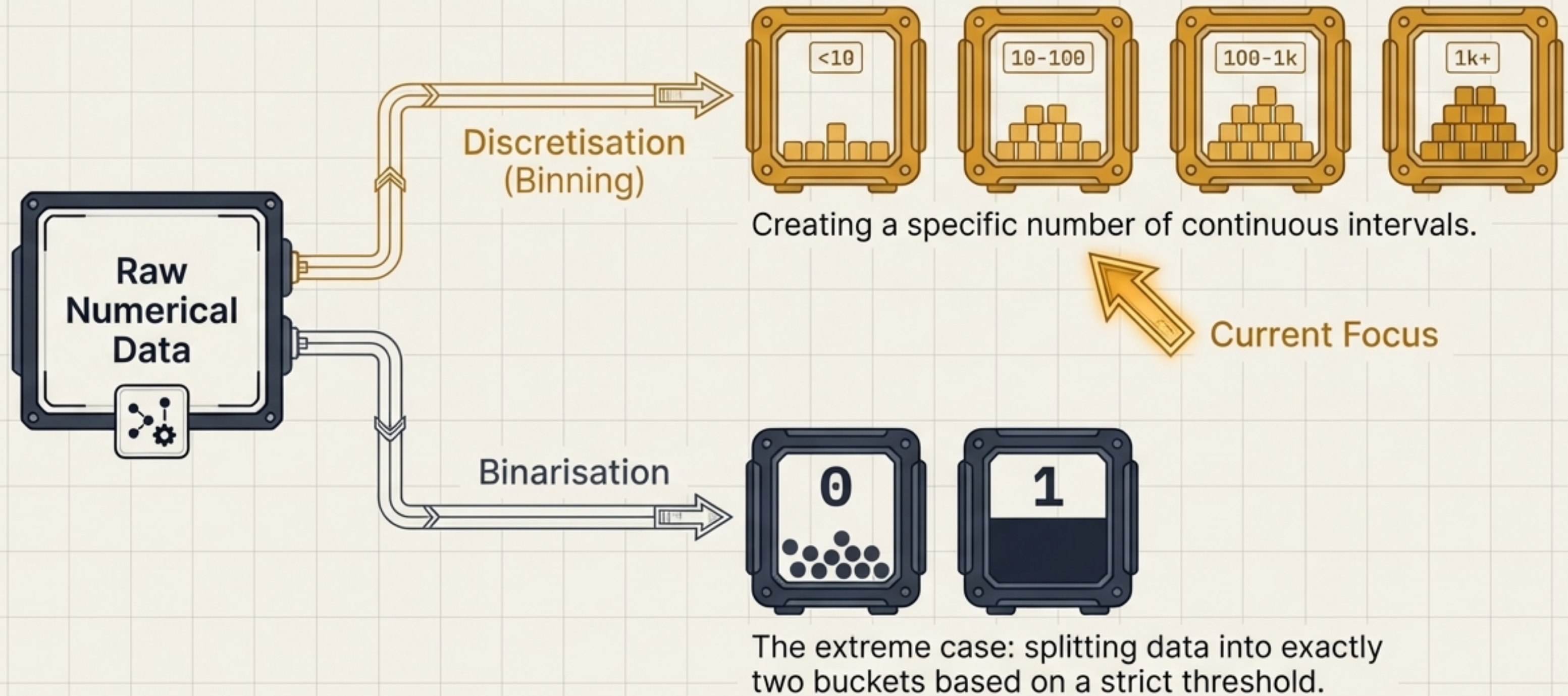


The Fix: Discretised Buckets

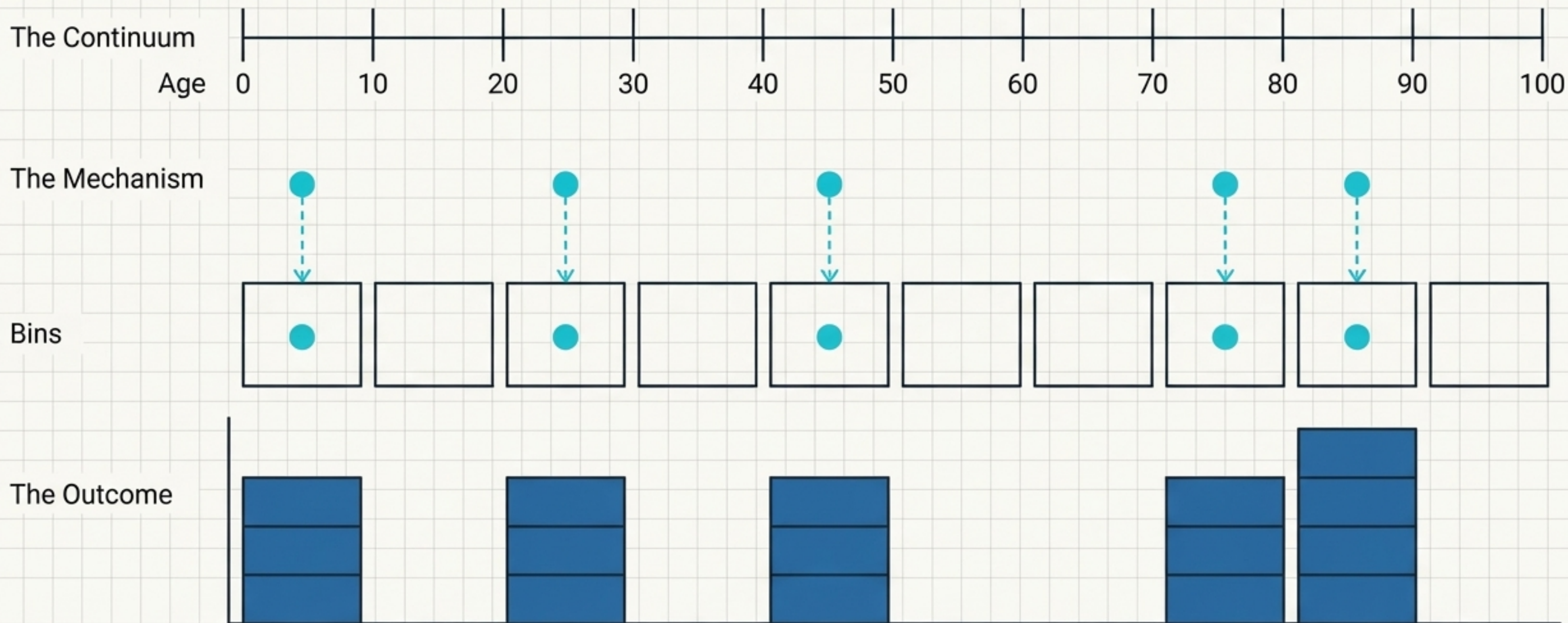


Continuous numbers are not always a model's best friend. Forcing extreme ranges into discrete categories tames outliers, reduces noise, and exposes non-linear relationships to the algorithm.

The Transformation Taxonomy



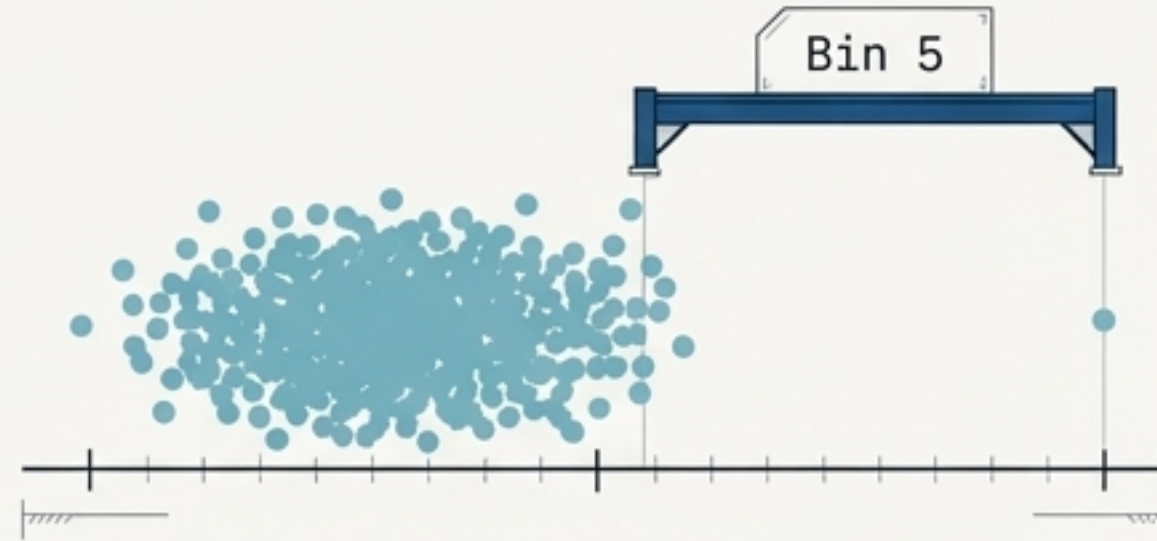
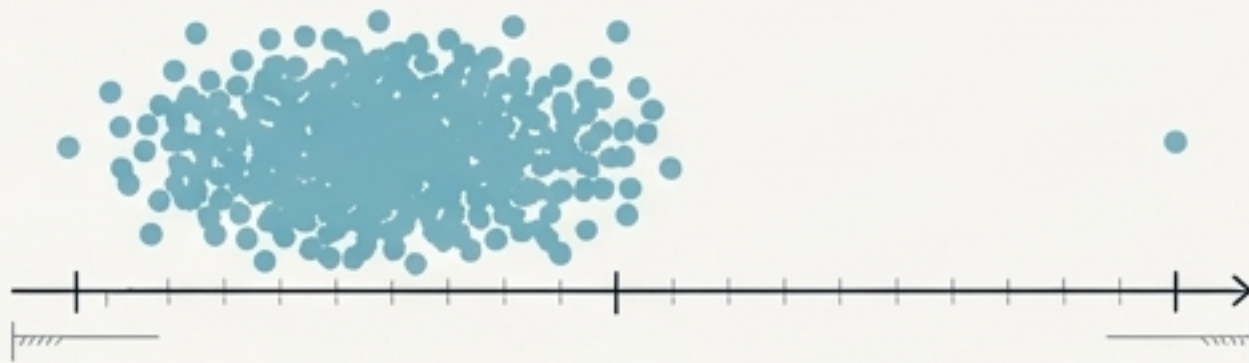
Deep Dive: What is Discretisation?



Every numerical observation is evaluated and permanently reassigned an integer representing its newly designated 'bin' identifier.

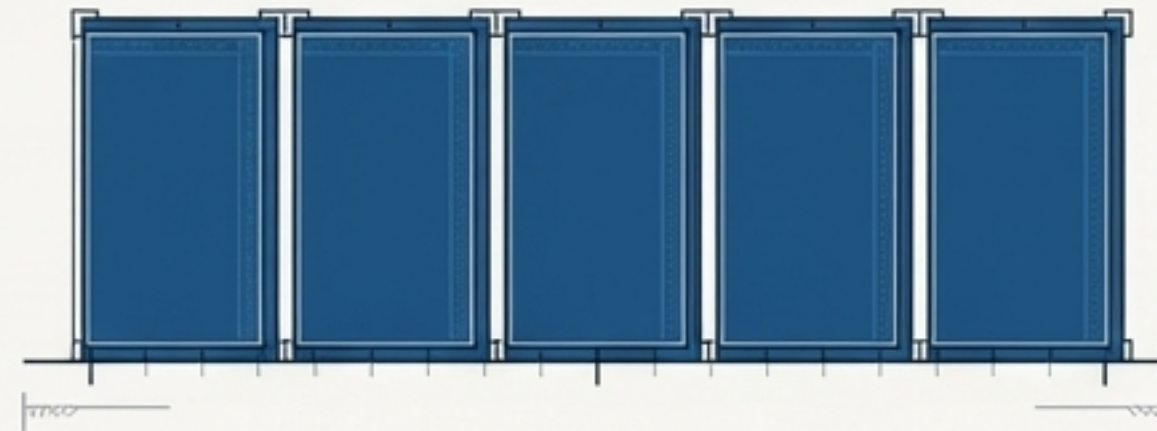
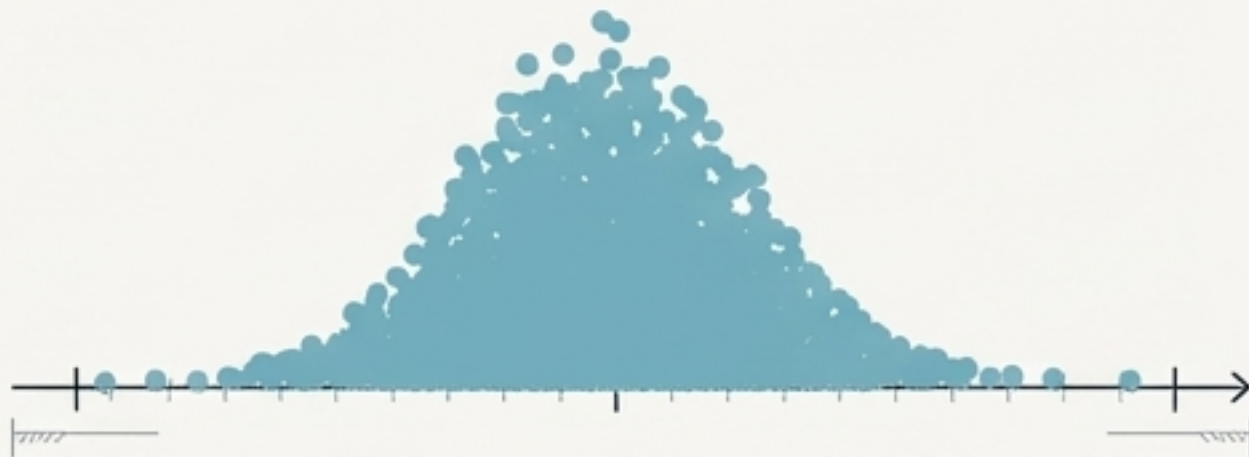
The Two Core Benefits of Binning

Benefit 1: Handling Outliers



The extreme outlier is structurally forced to be treated identically to the normal edge data.

Benefit 2: Improving Spread



Skewed data is manipulated to ensure a uniform distribution, providing the algorithm with a balanced view of the variance.

The Arsenal: Types of Binning

Unsupervised Binning



Algorithmic, math-driven logic operating blindly on the data structure.

- Uniform (Equal Width)
- Quantile (Equal Frequency)
- K-Means (Clustered)

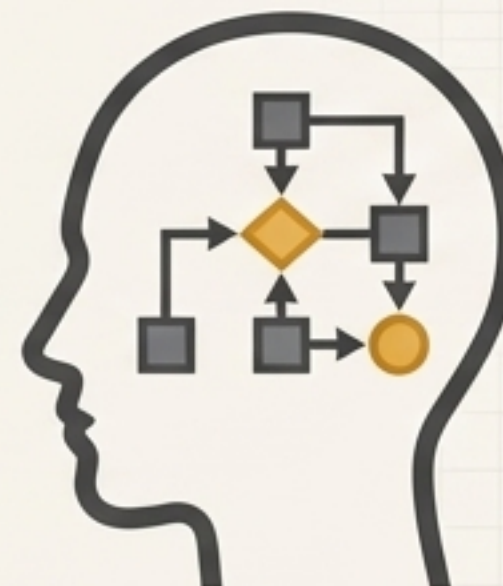
Supervised Binning



Target-aware sorting driven by the desired output variable.

*e.g., Decision Tree Binning
(Advanced: Out of Scope)*

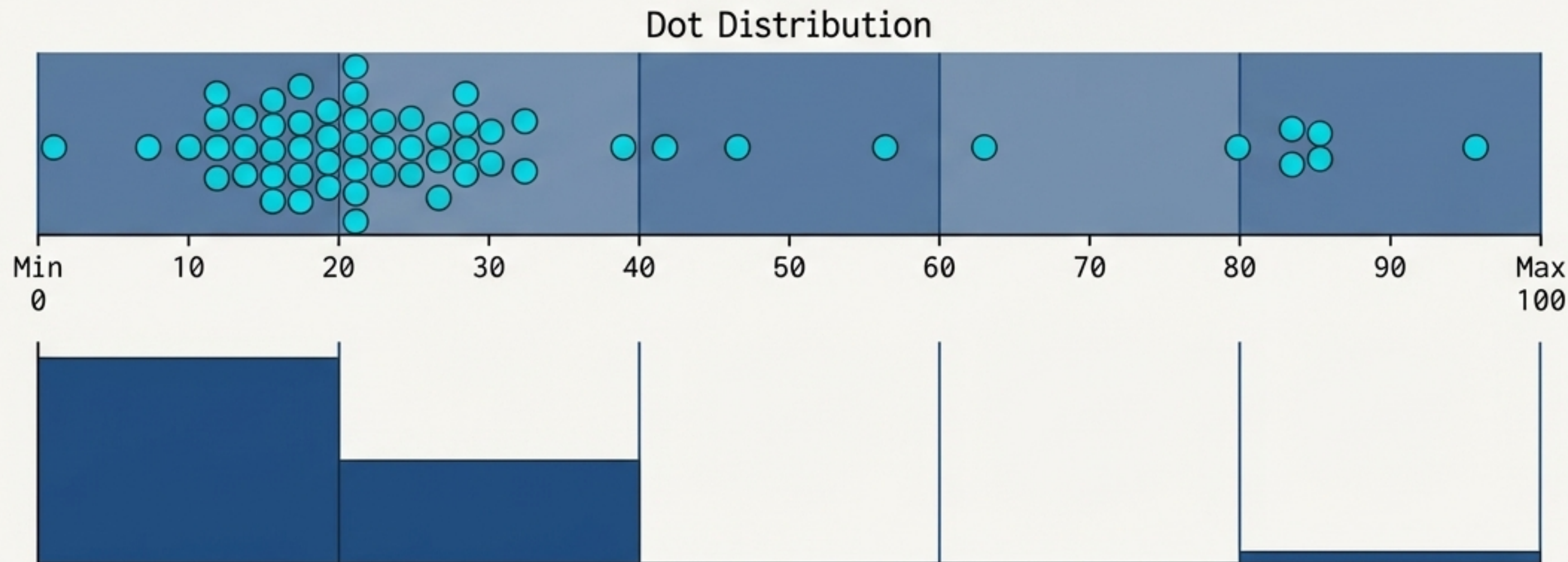
Custom Domain Binning



Human-guided intervals driven entirely by specific business contexts and domain knowledge.

Unsupervised Method 1: Uniform (Equal Width)

$$\text{Width} = \frac{(\text{Max} - \text{Min})}{\text{Number of Bins}}$$

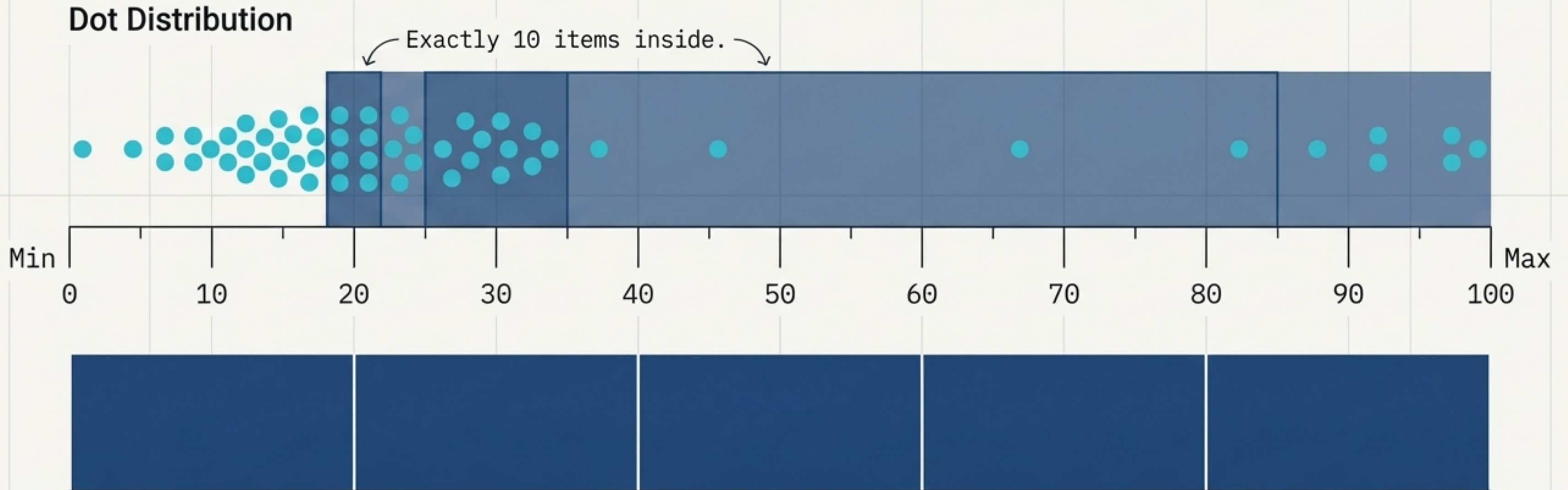


The physical bin widths are perfectly equal, but the frequency of data points within each bin varies wildly.

Outcome: Effectively catches outliers at the extreme ends, but fundamentally fails to change the underlying skewed spread of the data.

Unsupervised Method 2: Quantile (Equal Frequency)

Based on percentiles.
Every bin holds exact
equal statistical weight.

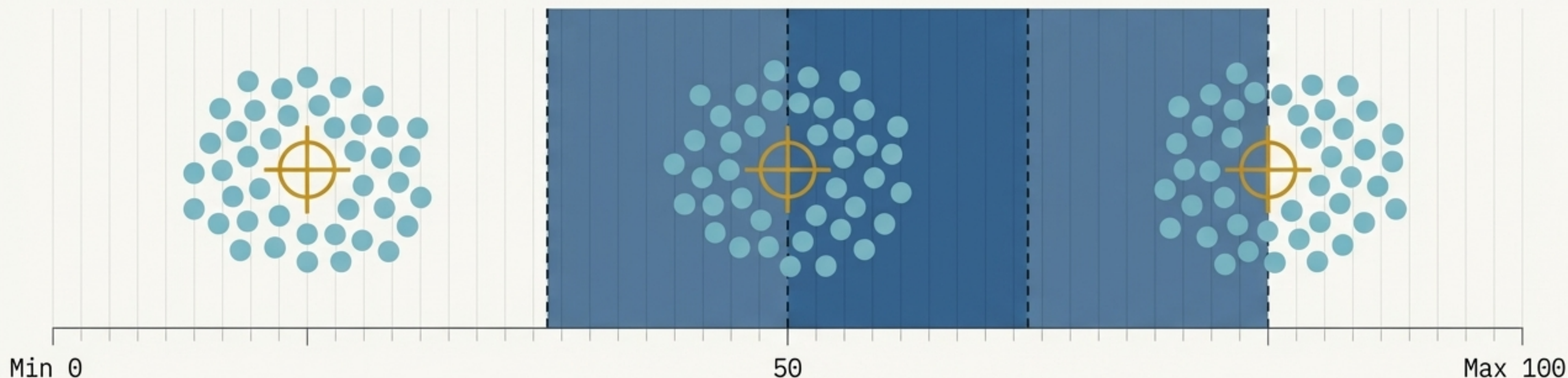


The Industry Default: Quantile binning simultaneously solves for extreme outliers whilst transforming skewed variance into a perfectly uniform data spread.

Unsupervised Method 3: K-Means (Clustered)

Iteratively locates natural groupings in multi-dimensional space.

Dot Distribution



Highly Specialised Application

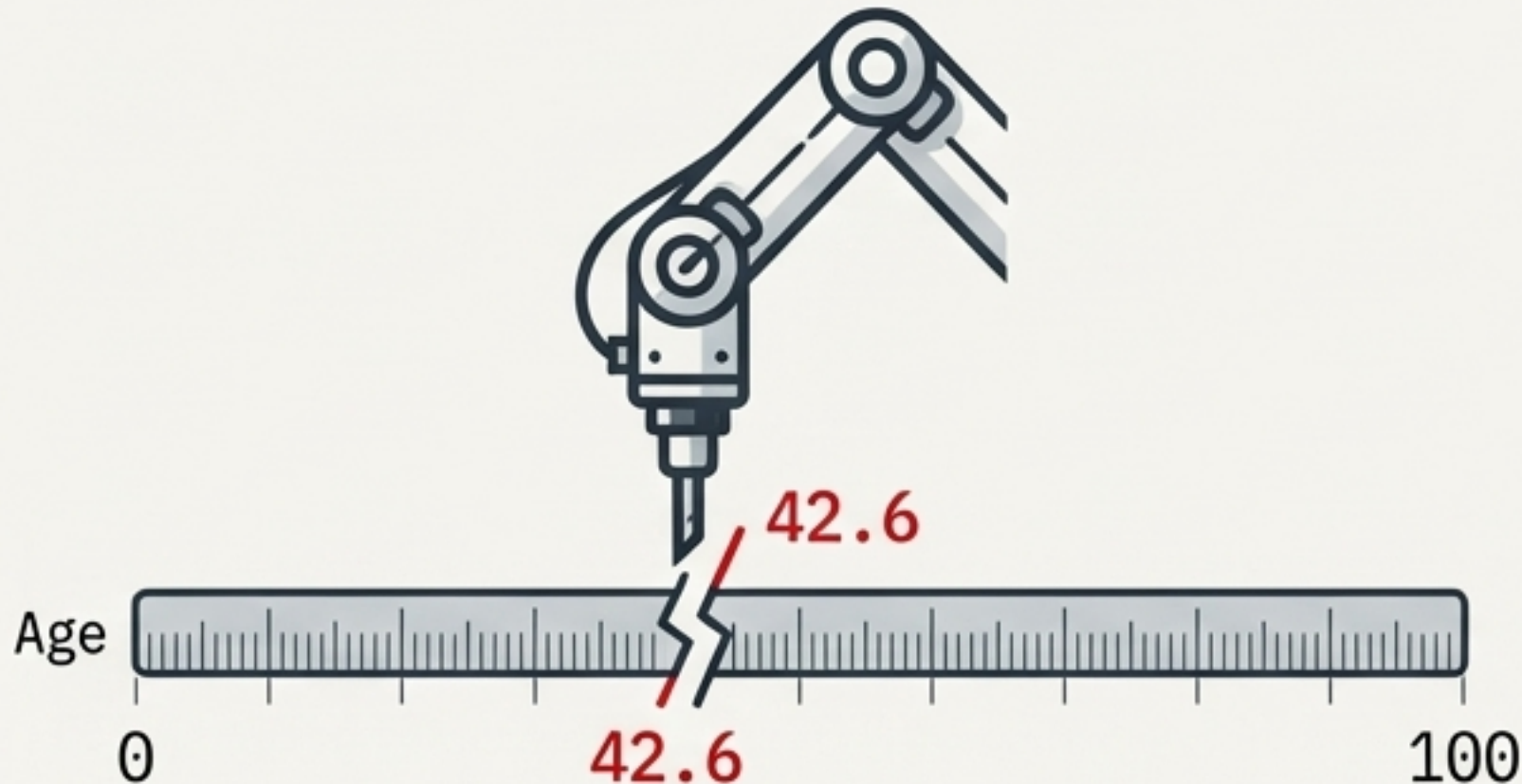
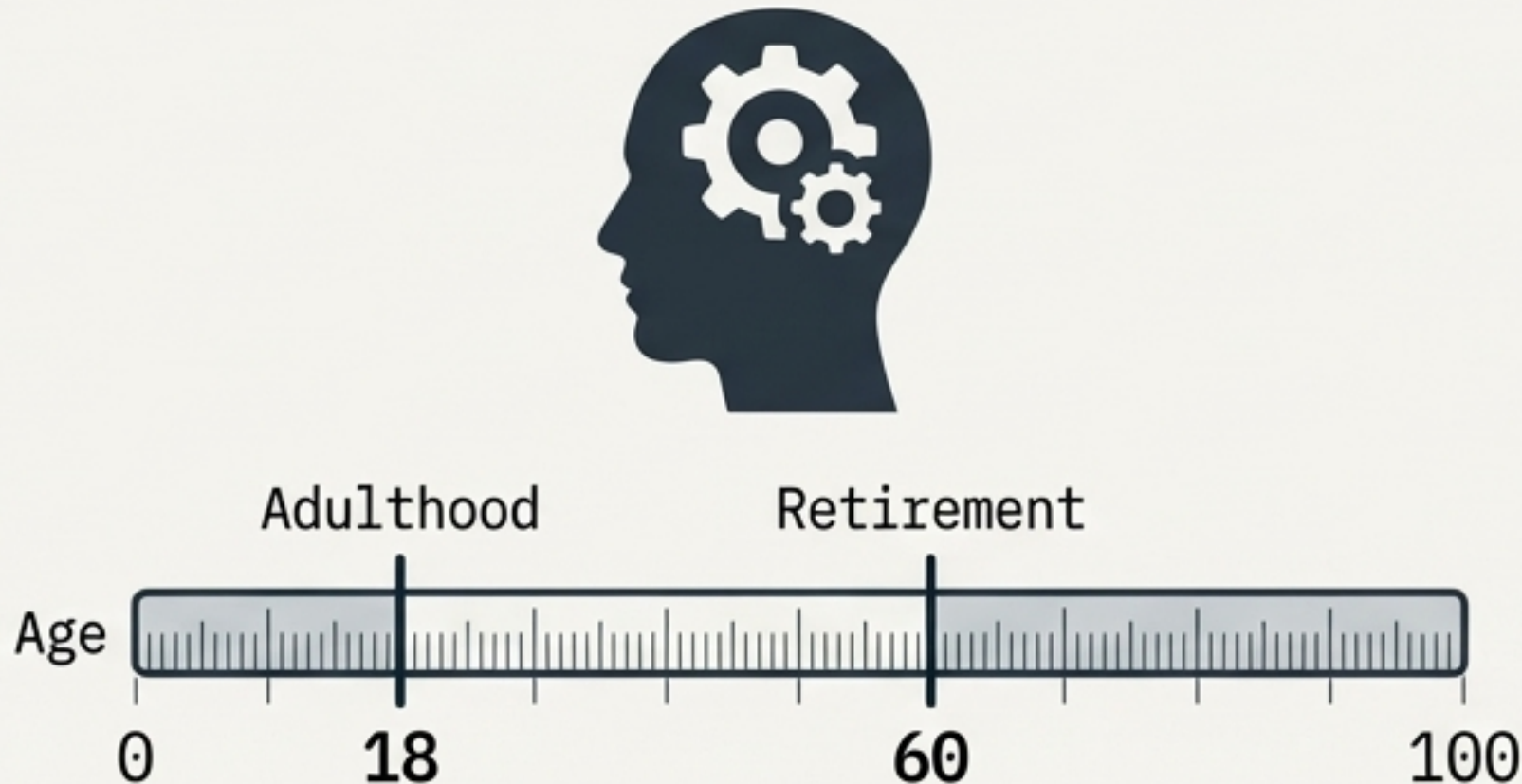
Only deploy K-Means binning when the raw data naturally forms distinct, separated clusters. If the data represents a smooth continuum without natural gaps, default to the Quantile strategy.

The Diagnostic Synthesis Matrix

A structural comparison of unsupervised discretisation algorithms.

Algorithm	Interval Logic	Effect on Data Spread	Best Use Case	Scikit-Learn Strategy
Uniform (Equal Width)	Fixed width math	No change	Simple baselines	<code>`strategy='uniform'`</code>
Quantile (Equal Frequency)	Percentile weight	Flattens to uniform	Default / Handling heavy skew	<code>`strategy='quantile'`</code>
K-Means (Clustered)	Centroid distance	Preserves natural clusters	Naturally clumped data	<code>`strategy='kmeans'`</code>

The Human Touch: Custom Domain Binning

Algorithmic Limitation	Human Context & Business Logic
 Unsupervised algorithms operate without human context, creating mathematically perfect but practically meaningless boundaries.	 Domain experts divide data based on real-world constraints and critical business thresholds.

❶ **Implementation Note:** Scikit-Learn's automatic discretisers cannot process custom human thresholds. Injecting true domain knowledge requires custom logic using Pandas (e.g., `pd.cut()`).

Implementation Blueprint: KBinsDiscretizer

```
from sklearn.preprocessing import KBinsDiscretizer  
discretizer = KBinsDiscretizer(n_bins=10, strategy='quantile', encode='ordinal')
```

1

2

3

Resolution:

Define how many distinct categories are required for the model. **10**.

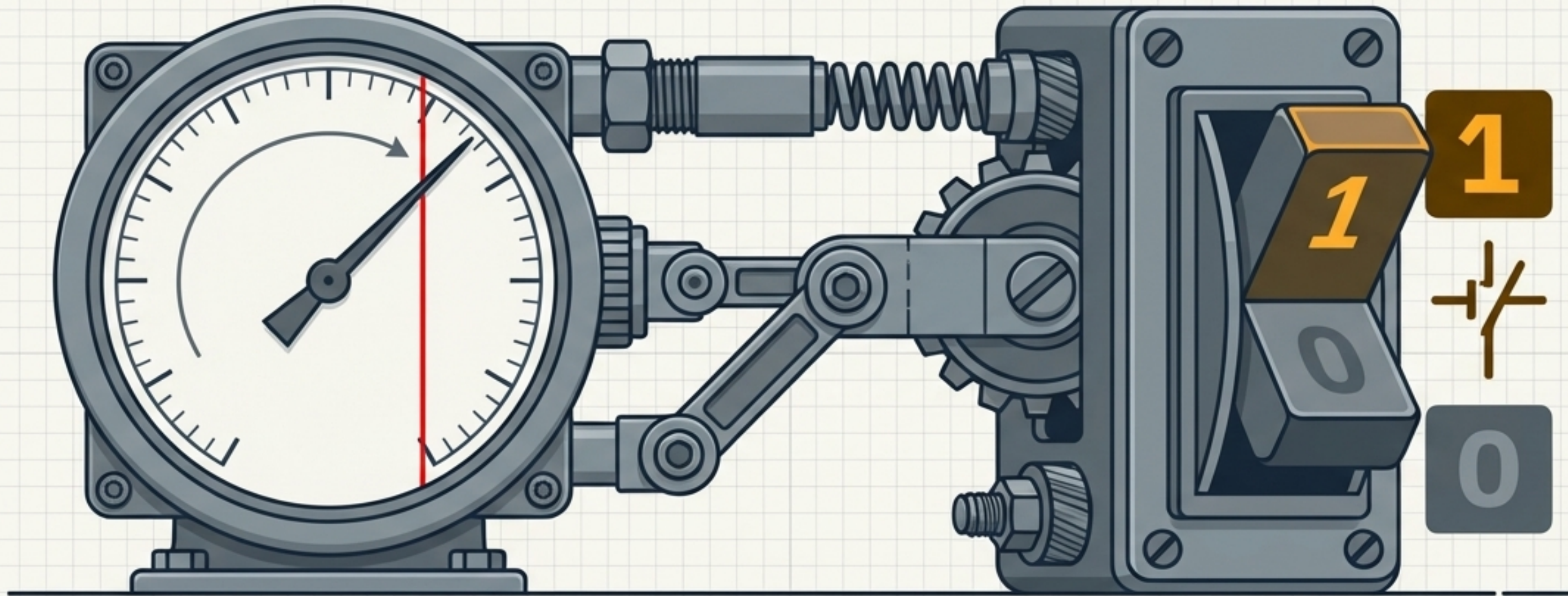
Mathematical Logic:

Choose the algorithmic approach (uniform, **quantile**, kmeans).

Output Architecture:

Determine the encoding format (**ordinal** integers vs. one-hot arrays).

Binarisation: The Extreme Discretisation



Transforming a continuous variable into exactly two definitive states based on a single, strict threshold. All underlying nuance is intentionally destroyed to mathematically amplify a specific binary signal.

Real-World Applications of Binarisation

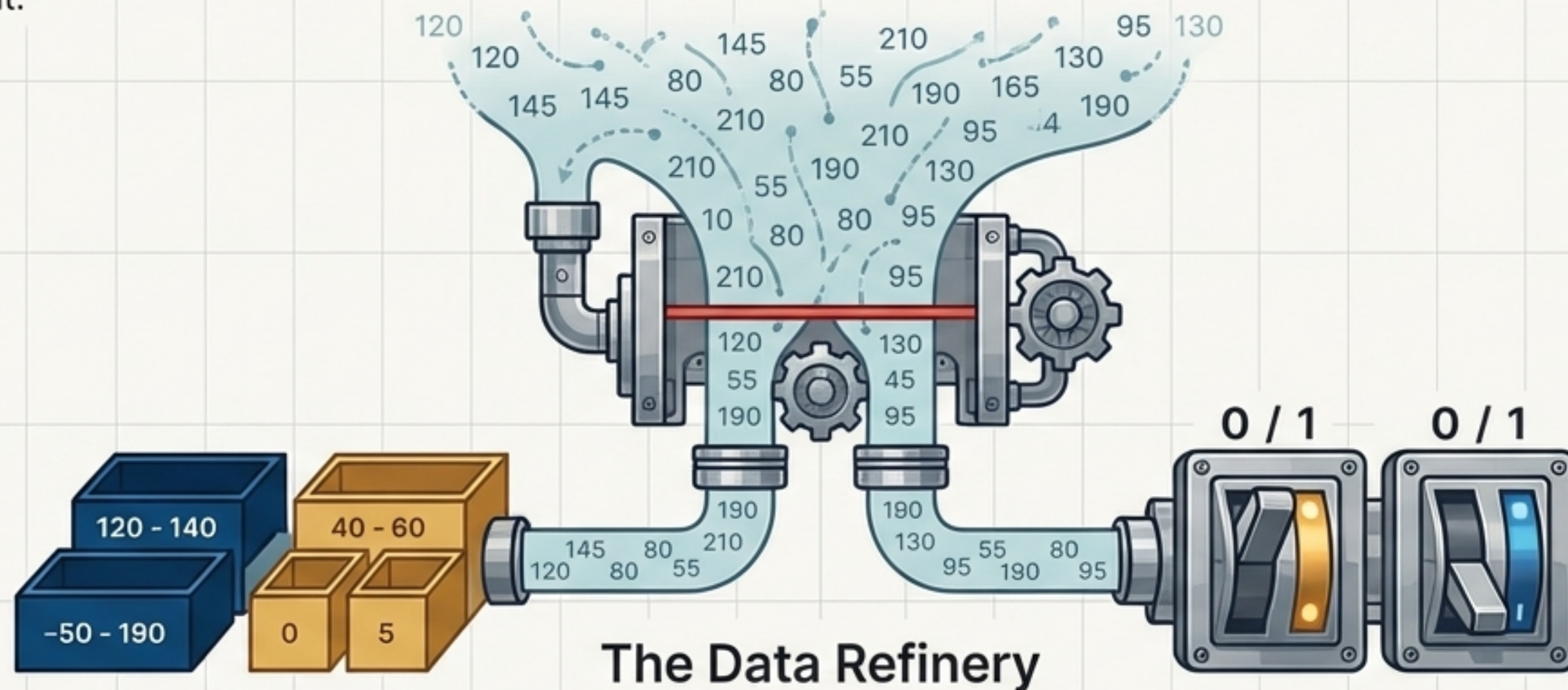
Data Refinery	Data Refinery	Data Refinery
Finance	Computer Vision	Behavioural Data
		
Annual Income Data Threshold: £60,000 (6 Lakhs)	Pixel intensity (0-255) Threshold: 127.5	Siblings + Parents aboard Threshold: 0
< £60k → Non-Taxable (0) > £60k → Taxable (1)	< 127.5 → Black Pixel (0) > 127.5 → White Pixel (1)	0 → Travelling Alone (1) > 0 → With Family (0)

Implementation Blueprint: Binarizer

The exact cut-off point defining the strict 0/1 split.

```
from sklearn.preprocessing import Binarizer  
binarizer = Binarizer(threshold=127.5, copy=False)
```

False aggressively overwrites existing data; True generates a new discrete feature.



Whether binning into groups or slicing into binary states, feature engineering is the art of strategic simplification—reducing noise to reveal the undeniable signals your models crave.